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Practical estimation of erosion damage caused by solid particle impact Part 2: Mechanical properties of materials directly associated with erosion damage

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Abstract

In order to select high erosion resistant materials or to estimate erosion damage caused by solid particle impact in actual cases, it is very important to take the mechanical properties of a material into account. It is well known that material hardness is closely related to erosion damage to materials. Another mechanical property apart from material hardness should be investigated, since the material hardness is not necessarily an essential indicator for industrially useful materials. From the fact that material removal is caused by indentation processes, material hardness is clearly one of the predominant parameter. It is also proposed that a new mechanical property directly associated with erosion damage exists in the indentation phenomenon. In quasi-static indentation tests using a WC ball, load relaxation after the completion of the indentation was measured and characterized for aluminum, copper, carbon steel and stainless steel specimens. It was found that the degree of load relaxation depended upon the ability of plastic flow for soft materials and upon brittleness accompanied by intergranular cracks between composite phases for hard metallic materials. It was concluded that a predictive equation containing both material hardness and the load relaxation ratio could be related to estimate experimental erosion damage data for many types of materials under a variety of impact conditions.
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Keywords: Estimation of erosion damage; Predictive equation; Solid particle impact; Mechanical properties; Hardness; Plastic flow; Brittleness; Metallic materials

1. Introduction

It is very important to develop predictive equations for erosion damage under a wide range of particle impact conditions in order to select highly erosion resistant materials or for estimating the life span of a material in service. A predictive equation was proposed in the previous paper [1] and the effects of impact parameters on impact angle, velocity and particle diameter dependence of erosion damage were investigated for three types of metallic materials. It was found that material hardness and particle properties are the predominant parameters that affect various dependences of erosion damage. It was however out of focus that material property dependence of erosion damage and effective parameters against its dependence. A number of research papers on material hardness [2–4] have presented over the past 10 years. It is generally concluded that the material hardness is exactly one of the mechanical properties associated with erosion damage, and valid in the cases of same compositional material systems and of a limited hardness range. However, Finnie reported that material hardness is independent of the amount of erosion in the case of heat-treated carbon steels [5]. Thus, material hardness is not necessarily related to erosion damage. Elastic modulus, yield strength and other properties are thought to affect erosion damage sometimes with the mixed effects of these properties [6,7]. Brittleness, which is one of the usual properties that seems to be useful for ceramic material and is possibly valid for metallic materials as well. These mechanical properties, except for

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hardness, can be usually obtained from tensile or fractural behavior. However, erosion is considered to be based on indentation behavior so that it is probably connected with mechanical properties related to indentation behavior.

In this paper, additional mechanical properties of materials different from material hardness were discussed to be composed into a predictive equation of erosion damage applied to any types of industrial materials under any impact conditions. Indentation tests were conducted to obtain other mechanical properties for a wide range of materials (Vicker's hardness number Hv: 0.4–8.0 GPa) using a universal testing machine. Load relaxation during an indentation process was noted to detect plastic deformation characteristics or the brittle nature of materials. Erosion tests were also performed for various materials. Attempts were made to correlate these properties with erosion damage at normal angle, E_{90} (volume loss calculated with mass loss divided by the density of the material per mass of erodent ($mm^3\,kg^{-1}$)). A predictive equation, which composed of all predominant parameters, was then proposed.

2. Experimental details

A sand blast type erosion test rig used in this study was described in the previous paper [8,9]. Erosion tests were performed under impact conditions of normal angle, 50–150 m s^{-1} and particles of SiO_2 and SiC (49–428 μm, mean diameter). Erosion rates at normal angle, E_{90} were obtained under impact conditions. The materials used in this study were three types of aluminum–copper, iron–carbon steels and stainless steels, as described in the previous paper [1]. The particles and impact conditions were also listed in the table in the paper [1].

A universal testing machine was used for quasi-static indentation tests with a 3 mm diameter WC ball, as shown in Fig. 1. The WC ball, used as an indenter, was attached at the tip of a load cell and depressed on a target material with a crosshead at a constant speed of 0.083 m s^{-1}. The ball was stopped when the load reached a certain value, and its position maintained for 300 s. Metallic material tends to relax the load after the completion of the indentation. The load behavior with testing time was measured by the load cell, which was connected to an A/D converter and a computer. The load cell is composed of the detecting bars of strain. As the load is detected as elastic strain of the bars on which strain gauges are mounted, the load cell has displacement during the indentation. The displacement of the load cell is generally ignored in tensile tests involving large displacement. However, in this case, the displacement of the crosshead is the sum of the minute displacement of the bar in the load cell and an indentation depth of the indenter into the target material. In order to obtain the minute displacement of the load cell, dummy intrusion tests were conducted using the same measurement system without the indenter. The true indentation depth was then obtained by subtracting the displacement of the load cell from that of the crosshead. On the other hand, the load indicated by the load cell was quite the same as that acted on the indentation because of the absence of effects of the elastically strained bar in the load cell.

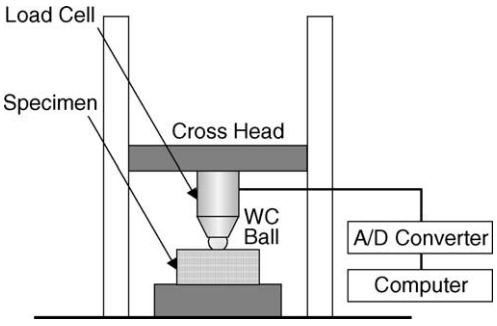


Fig. 1. Schematic diagram of a universal testing machine.

Dynamic indentation tests were performed using a gas gun testing unit [10]. A WC ball, of 3 mm in diameter, was impacted on the target materials at impact velocities from 50 to 130 m s^{-1} at normal angle. The crater diameter D, indentation depth δ and lip height h_L in units of mm were measured using an optical microscope or a surface proilometer to compare with those for the quasi-static indentations.

Cross-sectional surfaces beneath and far from the indentations in the target materials were observed by scanning electron microscopy (SEM), to investigate effects of indentation on the microstructure.

3. Results

Fig. 2 shows the relationships between material hardness and erosion damage at normal angle, E_{90} ($mm^3\,kg^{-1}$) at impact velocities of 104 and 121 m s^{-1} with SiO_2 with a mean diameter of 326 μm. The relationships were different among the three compositional material systems of

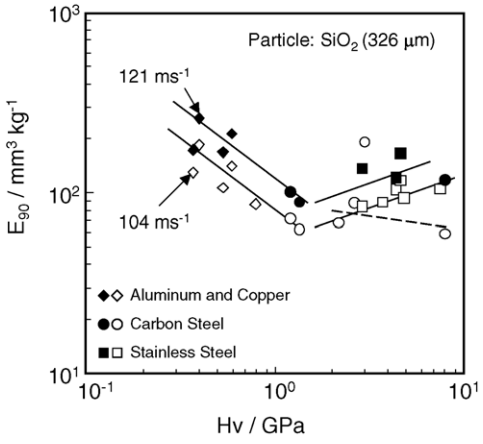


Fig. 2. Relationships between material hardness and E_{90} at impact velocities of 104 and 121 m s^{-1} with 326 μm SiO_2.

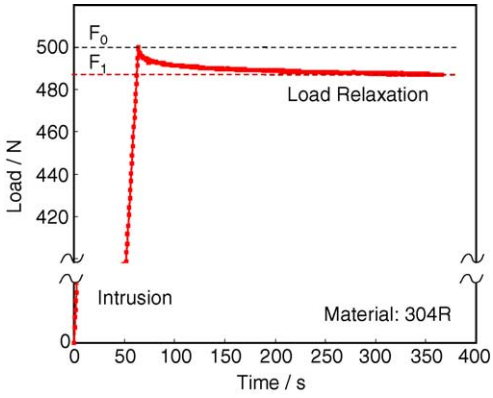


Fig. 3. The load signal-testing time curve for type 304R caused by the intrusion of a 3 mm WC ball.

aluminum–copper, iron–carbon steels and stainless steels. The value of E_{90} decreased with an increase in material hardness for the aluminum–copper specimens, but increased for the stainless steels. The E_{90} for iron–carbon steels, some of which were made by heat treatments, had nearly the same values. The erosion damage was increased with increasing impact velocity, but the slopes on a logarithmic scale were nearly same for the two impact velocities.

Fig. 3 shows an example of the load signal-testing time curve for type 304R stainless steel caused by the intrusion of a WC ball. The load increased with testing time during the intrusion process and next gradually decreased as the load was relaxed, after the completion of the indentation. The ratio of the load of the relaxation was defined by $(F_0 - F_1)$ divided by F_0 in unit of N. The intrusion and relaxation behaviors depended upon the type of materials. Fig. 4 shows examples of the relations between contact pressure–strain (ε) curves for 304R and FER specimens. The indentation depth was equal to the modified displacement converted from the speed of the crosshead in Fig. 3, and the indentation ratio d/D was calculated from the indentation depth. The strain was introduced by Tabor [11] as $\varepsilon = 0.2d/D$. The contact pressure was obtained by dividing the detected load by the contact area, which was geometrically calculated from the indentation depth. The

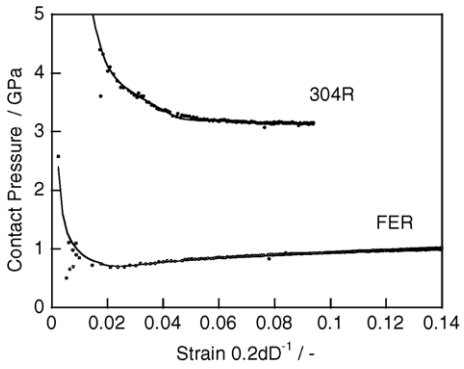


Fig. 4. Contact pressure-strain curves for 304R and FER specimens.

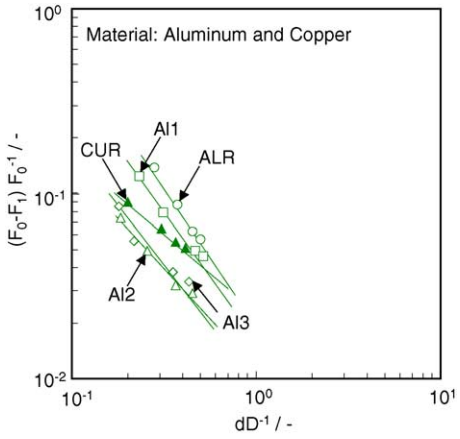


Fig. 5. Relationships between the load relaxation ratio and indentation ratio for the aluminum–copper system.

contact pressure quickly dropped with scattering of the data and then gradually shifted with an increase in strain. The terminal contact pressure was nearly same as the Vicker's hardness number (GPa).

Figs. 5–7, respectively, show the relationships between the load relaxation ratio $(F_0 - F_1)/F_0$, and indentation ratio d/D for the three compositional material systems. The behavior of the load relaxation ratio was roughly linear on a logarithmic scale. The slopes for all of the aluminum specimens were negative and larger than those for the other materials. The slopes for the carbon steel specimens were generally small compared with aluminum and a few specimens of stainless steel. The slopes for the stainless steel specimens were scattered over a wide range.

Fig. 8 shows examples of the difference in lip height ratio h_L/d to indentation depth ratio δ/R (R: radius of an indenter) between the quasi-static and dynamic indentation behaviors. The lip height ratio in the dynamic indentation was larger than that in the quasi-static one for a similar indentation depth ratio δ/R for type 304R. In Fig. 8(a) the slopes in the quasi-static and dynamic indentation behaviors were the same in

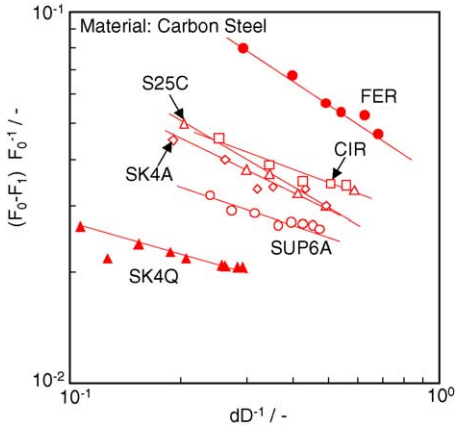


Fig. 6. Relationships between the load relaxation ratio and indentation ratio for the iron–carbon steel system.

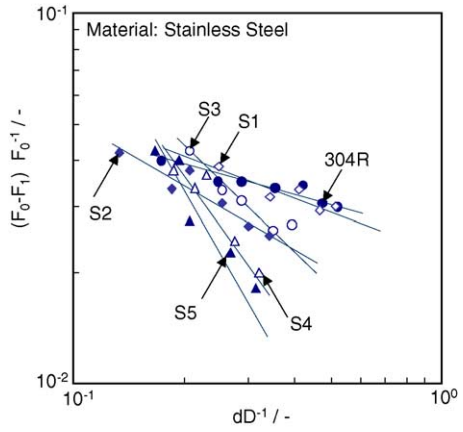


Fig. 7. Relationships between the load relaxation ratio and indentation ratio for the stainless steel system.

the case of stainless steel of type 304R. Many of the materials tested followed this type of h_L/d behavior. On the other hand, Fig. 8(b) shows examples of the different slopes between the two indentation processes in the case of cast iron as received CIR, stainless steels S2 and S3.

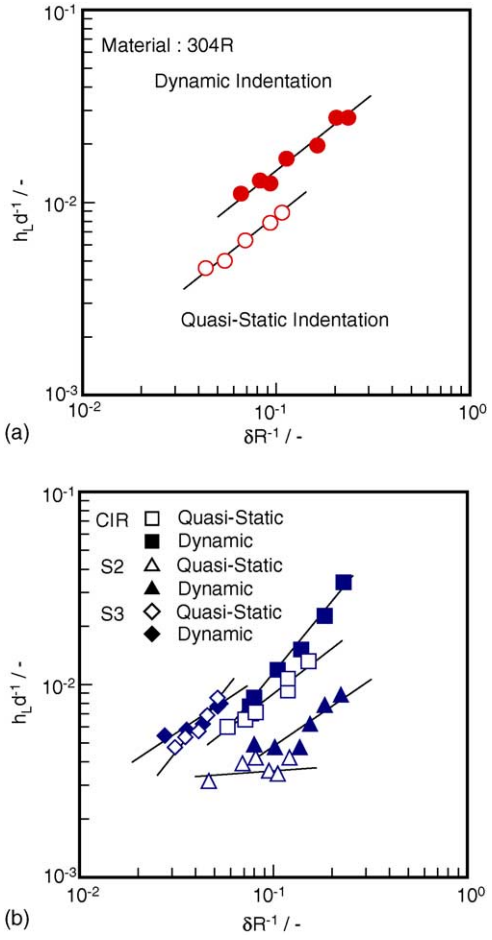


Fig. 8. Comparison between the quasi-static and dynamic indentation behaviors indicating relations between the lip height ratio and the indentation depth ratio.

4. Discussion

4.1. Relationship between E_{90} and material hardness

It has been recognized over the past 10 years that erosion resistance is directly related to the hardness of pure metals [3] or materials of the same composition [2] and is not related in the case of heat-treated carbon steels [5]. The results in Fig. 2 were in good agreement with previous results. The hardness in Fig. 2 is the initial hardness of a non-eroded surface. Curious test results were obtained in the stainless steel materials, that is E_{90} increased with an increase in hardness with the scattered data. These results suggest that erosion damage depends, not only on material hardness, but also other mechanical property or metallic microstructures.

For example, in this study we cite the iron–carbon steel system which is composed of pure iron, mild steel, spring steel and two of heat-treated carbon steel with the same carbon content, but microstructures that appeared on the surface are single or composites with quite different structures such as ferrite, pearlite and martensite. The microstructures in the stainless steel system are also very complicated, and are from a single into composite structures. Erosion damage may be related to material hardness, if we select materials, which have a different ratio of microstructures. The relation among materials in which the microstructures are ignored may be meaningless in terms of the material hardness. It is therefore necessary to take into account other mechanical properties, which are related to microstructures and, indirectly, with material hardness, if the relation of erosion damage against all industrial materials is to be considered.

The effects of impact velocity on slope in Fig. 2 were not recognized in these experiments. This is inconsistent with the small effect of material hardness on the impact velocity dependence found in the previous paper [1]. This was probably visible by the scattering of data depending on the material hardness. The slope should be regarded to be independent of impact velocity taking account of advantages for predicting erosion damage.

4.2. Work-hardening during indentation processes

The contact pressure behavior in Fig. 4 can be regarded as indentation resistance or a type of plastic flow of material [11] and as Brinell hardness at the terminal point. The plastic flow behavior in the indentation process after the minimum contact pressure point ($\varepsilon = 0.02$) is also considered to simulate the ability of work-hardening. This strain is large enough to the strain at the yield point due to an ordinary yield strain value of 0.002. We tried to adopt the ability of work-hardening but were not able to extract an exponent number because nonlinear curves and negative work-hardening were sometimes observed as in the case of type 304R as shown in Fig. 4. We noted another mechanical property from the indentation behavior.

4.3. Load relaxation ratio and correlation of E_{90}

The load relaxation ratio can be considered to indicate one of the mechanical properties of a material. It is generally considered that load relaxation increases with an increase in load to the size of the indentation. This indicates the possibility of a constant load relaxation ratio against the increased indentation ratio. However, for all of the materials used in this study the load relaxation ratio decreased with the indentation ratio. The decrease in the ratio is associated somehow with work-hardening or an unknown mechanical property of the material. The load relaxation of material is due to delayed plastic deformation that occurs after the completion of the indentation, which cannot adequately respond during the indentation process.

Fig. 9 shows examples of SEM observations of etched cross-sectional surfaces at a distance from the indentation (a), (c) and (e) and near the bottom area of the indentation (b), (d) and (f) for the S2, S3 and S5 specimens. It can be seen that the brittle iron and chromium boride grains in the austenitic matrix are fractured by compressive stress for S2 (b). The S3 specimen had microstructures with small grain sizes and had little cracks on the cross-sectional surface (d). In the case of the S5 specimen, intergranular cracks of martensitic grains appeared by the indentation (f). The decrease in load relaxation ratio was promoted by the generation of intergranular cracks (S5) or the destruction of brittle microstructures (S2) in the case of harder materials. The negatively large slope for the load relaxation therefore indicates the brittle characteristics of the materials. On the other hand, no cracks could be observed in the aluminum specimen, in which a large decrease in load relaxation ratio was found. In the case of aluminum, the decrease in relaxation with indentation ratio is considered to be due, not to intergranular crack propagation, but to adequate plastic deformation during the indentation process. The cause of the load relaxation is, however, quite different between softer and harder materials.

The behavior of the load relaxation ratio can be approximated by Eq. (1).

$$\frac{F_0 - F_1}{F_0} = a\left(\frac{d}{D}\right)^{-b} \tag{1}$$

where a and b are a constant and an exponent, respectively. Fig. 10 shows the relation between b and E_{90} at impact velocities of 99 m s^{-1} for SiC and 104 m s^{-1} for SiO_2. It is found that the correlation of E_{90} was relatively improved when the load relaxation ratio for the three types of materials, although the data were scattered for the materials. We cannot comprehend the reason why erosion damage is related only to load relaxation and we attempted to determine a reasonable relation based on Eq. (1) using the hardness and load relaxation

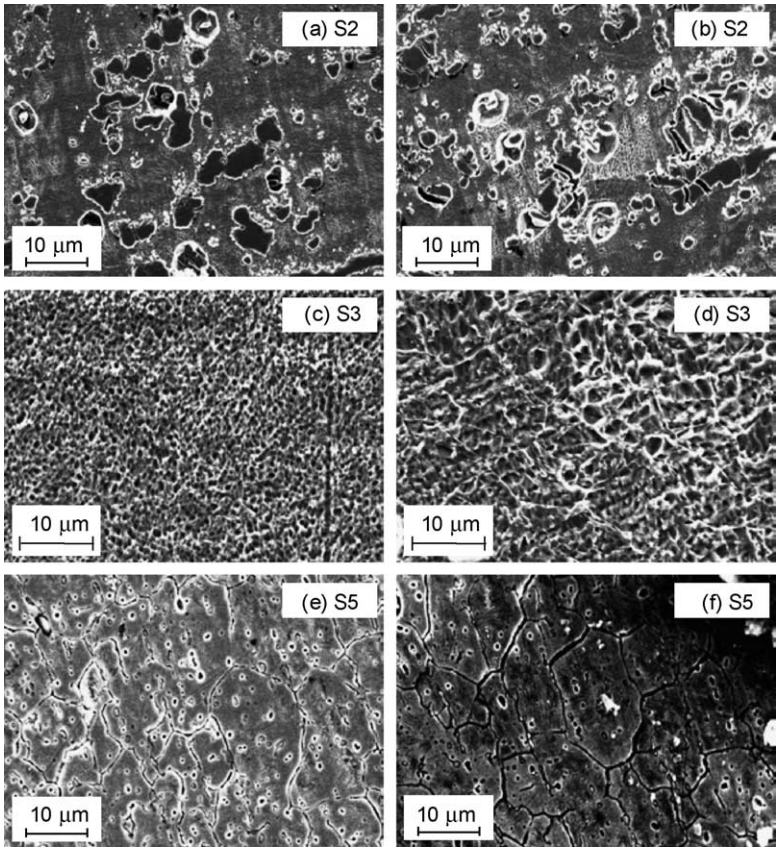


Fig. 9. SEM observations of cross-sectional surfaces for S2 (a and b), S3 (c and d), S5 (e and f) specimens.

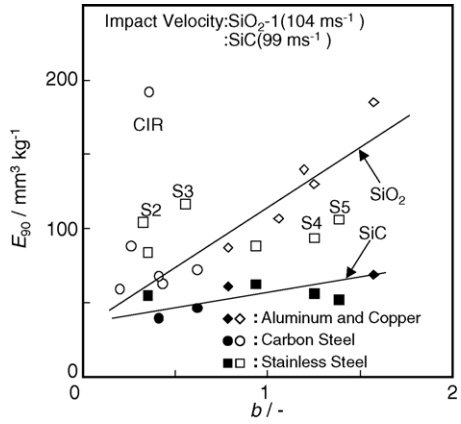


Fig. 10. Relations between b and E_{90} at impact velocities of 99 m s^{-1} (SiC) and 104 m s^{-1} (SiO_2).

ratio. As a result of trial and error we propose the following correlative equation:

$$E_{90} = K(a\mathrm{Hv})^{-bk_1}, \quad (0 \le a\mathrm{Hv} \le 1, b \ge 0) \tag{2}$$

where a and b are the same as in Eq. (1). K and k_1 are a constant and an exponent, which are determined by the type of a particle used. Hv is an initial hardness number in units of GPa. In Eq. (2), E_{90} decreases with an increase in material hardness. E_{90} increases with an increase in b, which implies ability of plastic deformation or brittleness, if a material has the same hardness value. Therefore, Eq. (2) is consistent with the general concept of erosion damage, as defined by the hardness and brittleness of a material. Eq. (2) indicates that both the hardness and brittleness of a material in a complicated manner. The correlation of E_{90} by Eq. (2) is shown in Fig. 11. The correlation was improved considerably by Eq. (2), but three materials of CIR, S2 and S3 deviated from the correlation.

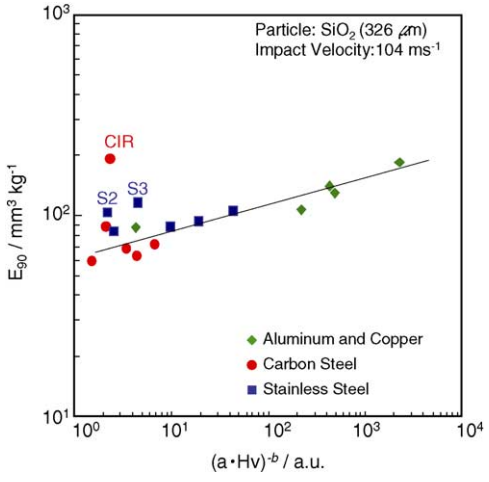


Fig. 11. Correlation between $(a\mathrm{Hv})^{-b}$ and E_{90} at an impact velocity of 104 m s^{-1} with 326 μm SiO_2.

4.4. Consideration of deviated materials from the correlation

From Fig. 8 it can be found that the quasi-static lip height behaviors of the materials deviated from the correlation of E_{90} were different from the dynamic behaviors. Both the material hardness and the load relaxation ratio are in quasi-static indentation behavior. However, the material removal by erosion is dynamic and, as a result, erosion damage should be essentially related to the dynamic properties of the material. This suggests that the good correlation between erosion damage at normal impact (E_{90}), and quasi-static properties of the hardness and the load relaxation ratio can be obtained for materials, which have similar quasi-statically and dynamically mechanical properties. The materials that deviated from the correlation were eventually CIR, SK4Q, S2 and S3. These materials had characteristic microstructures, examples of which are shown in Fig. 9(a)–(d) or a different behavior between the quasi-static and dynamic indentations, as shown in Fig. 8(b). SK4Q and S3 had fine microstructures or grains about a few microns of martensite, the size of which is comparable to the indentation caused by the impact of a particle with a few hundred microns in diameter. The erosion damage may be accelerated by the very brittle iron or chromium boride grains, which are very weak for the impact and were probably accelerated by the preferential removal of the brittle microstructure. CIR had the brittle and fine microstructure of graphite so that the erosion damage to CIR was considered to be largely accelerated by the mutual effect of the size and strength of the microstructure.

4.5. Establishment of predictive equations of erosion damage

The effects of impact parameters on the various dependences of erosion damage have been discussed in the previous paper [1]. The predictive equations for erosion damage, which can be utilized under any impact angle, velocity, particle size and any types of materials, are as follows:

$$E(\alpha) = g(\alpha)E_{90} \tag{3}$$

$$E_{90} = K(a\mathrm{Hv})^{k_1 b}\left(\frac{v}{v'}\right)^{k_2}\left(\frac{D}{D'}\right)^{k_3} \tag{4}$$

where $E(\alpha)$ is the erosion damage (mm^3 kg^{-1}) at an arbitrary impact angle and $g(\alpha)$ is the impact angle dependence of normalized erosion. K, k_1, k_3 are constant and exponents which take an arbitrary unit and are determined by the properties of the particle (type of particle) and k_2 is an exponent determined both by the material hardness and by particle properties, which were extensively discussed in the paper [1]. v and D are the impact velocity (m s^{-1}) and particle diameter (μm), respectively, and v' and D' are the standard impact velocity and the particle diameter which are used in the experiments for the correlations of erosion damage. The exponents of impact velocity and particle diameter dependences which were

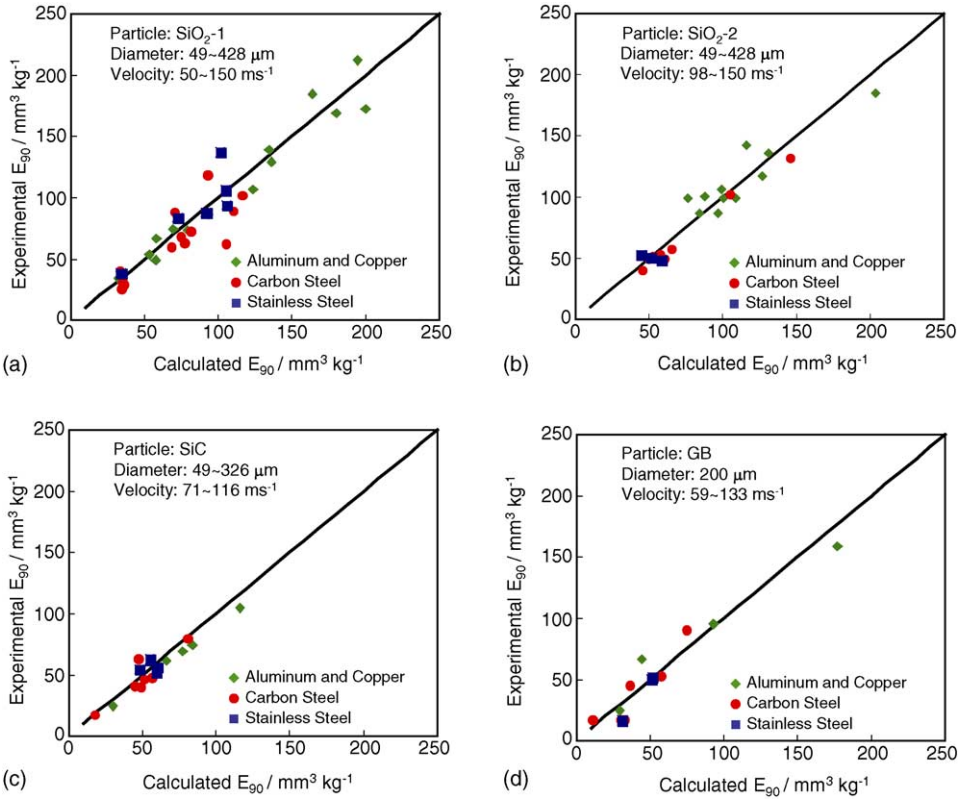


Fig. 12. Relationships between calculated E_{90} and experimental E_{90} for (a) SiO_2-1, (b) SiO_2-2, (c) SiC and (d) GB.

experimentally obtained can be also utilized in this equation using the dimensionless terms of (v/v') and (D/D'). The correlations between the values of E_{90} obtained experimentally and E_{90} calculated by Eq. (4) are shown for the three types of particles in Fig. 12. The materials, which have quite different mechanical properties between the quasi-static and dynamic indentation behaviors were eliminated from Fig. 12. The constants and exponents are listed in Table 1. It was considered that the constants and exponents of K, k_1, k_3, especially k_2 take individual values based on the type of particle, since the particle property which includes such as particle shape, angularity and hardness is not correlated with the impact conditions and other factors.

It can be seen in Fig. 12 that the calculated E_{90} is consistent with the experimentally obtained E_{90} for a wide range of the target materials, impact conditions for the respective type of particle and that the predictive equations proposed in this study are useful in practical service. It is anticipated that the complicated properties of really handled particles and erosion that is synergistically accelerated by corrosion from environmental species would make such an estimation difficult. It is very important to estimate K, k_1, which are reflected by the unknown particle property, taking account of the characteristic values of the known particle properties in Table 1. A practical estimation of absolute erosion damage can be completed by measurement of the mechanical property a, b in the load relaxation ratio and the initial hardness of the material according to Eq. (4). If the value of E_{90} can be obtained, erosion damage at an arbitrary impact angle $E(\alpha)$ can be estimated by Eq. (3), which was explained in the paper [1].

Table 1
Constants and exponents in the predictive equation

Particle	K	k_1	k_2	k_3	v' (m s^{-1})	D' (μm)
SiO_2-1	65	−0.12	$2.3(Hv)^{0.038}$	0.19	104	326
SiO_2-2	50	−0.12	$2.3(Hv)^{0.038}$	0.19	104	326
SiC	45	−0.05	$3.0(Hv)^{0.085}$	0.19	99	326
GB	27	−0.16	2.1	0.19	100	200

5. Conclusions

Other mechanical properties apart from material hardness were discussed and predictive equations for estimating erosion damage were established for a variety of industrial materials and under any impact conditions. The conclusions are:

1. The behavior of the load relaxation ratio depended upon the type of material and the extent of plastic deformation or brittleness.
2. The correlations of E_{90} with the characteristic constant and exponent obtained from the load relaxation behavior were excellent under a variety of impact conditions. It was found that erosion damage is associated with initial hardness and the plastic deformation characteristics or brittleness of a material.

3. Materials, which deviated from the correlation of E_{90} showed different behaviors between the quasi-static and dynamic indentations or characteristic microstructures and in these cases, erosion damage was usually higher than the calculated value.
4. The absolute erosion damage in a practical case can be obtained by estimating K, k_1, k_2 and k_3 values as particle properties according to the predictive equations proposed in this study. These predictive equations can be utilized for many materials and any impact conditions.

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